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$$\int_0^{1-} \operatorname{Bgsin} x dx$$

$$\Rightarrow \int \operatorname{Bgsin} x dx$$

$$\begin{cases} u = \operatorname{Bgsin} x \\ dv = dx \end{cases} \Rightarrow \begin{cases} du = \dfrac{1}{\sqrt{1-x^2}} dx \\ v = x \end{cases}$$

$$= x \operatorname{Bgsin} x - \int \frac{x}{\sqrt{1-x^2}} dx$$

$$= x \operatorname{Bgsin} x + \frac{1}{2} \int \frac{d(1-x^2)}{\sqrt{1-x^2}}$$

$$= x \operatorname{Bgsin} x + \sqrt{1-x^2} + C$$

$$\Rightarrow \int_0^{1-} \operatorname{Bgsin} x dx = \left[x \operatorname{Bgsin} x + \sqrt{1-x^2} \right]_0^1$$

$$= \operatorname{Bgsin}(1) - 1 = \frac{\pi}{2} - 1$$

References